helix_code_services.md — Services Guide

Revision: 2 **Last modified:** 2026-09-03T00:00:00Z

Overview

HelixCode service management wrappers: install, boot, stop, health-check, and status for the HelixCode / HelixLLM / HelixAgent service fleet.

Prerequisites

- `helix_code_home.sh` sourced first (provides `hc_home()`)
- bash ≥ 4.0
- Docker or Podman (for compose-based services)
- Network access to `github.com:HelixDevelopment/HelixCode.git`

Quick Start

```
source constitution/scripts/helix_code/helix_code_home.sh
source constitution/scripts/helix_code/helix_code_services.sh

# Install binaries
hc_install

# Boot the coder (multi-track live sink)
hc_boot_coder

# Check status
hc_status

# Verify all binaries on PATH
hc_verify_binaries
```

Functions

Function	Purpose
`hc_install [--flags]`	Install HelixCode binaries via `install_helix_path.sh`
`hc_boot_service <name>`	Boot a service via the Containers submodule compose
`hc_boot_coder`	Boot the HelixLLM coder on `:18434` (with health wait)
`hc_stop_service <name>`	Stop a service
`hc_status`	Print health status of every service in the `hc__status_endpoints` table
`hc_verify_binaries`	Verify all 6 binaries are on PATH
`hc_health [host] [port] [path]`	Health check over `http` only (from `helix_code_home.sh`)
`hc_health_url <url>`	Scheme-aware health check; prints `OK` or `DOWN`

Service Ports

These are the rows of the `hc__status_endpoints` table in `helix_code_services.sh` — that table is the single source, this table mirrors it. Verified against the consuming project's `scripts/systemd/*.service` units and a live `ss -ltnp` on 2026-09-03.

Service	Scheme	Port	Health Endpoint
HelixCode server	`http`	`:8080`	`/health`
HelixLLM coder	`http`	`:18434`	`/health`
HelixLLM gateway	`https`	`:8443`	`/internal/health`
HelixAgent	`http`	`:7061`	`/v1/models`
LLMsVerifier	`http`	`:8100`	`/health`

Notes:

- **HelixAgent is `:7061`, not `:8100`.** `:8100` is the **LLMsVerifier's** port; a prior revision of this table and of `hc_status` attributed it to HelixAgent, so `hc_status` reported "HelixAgent: DOWN" while HelixAgent was serving normally. HelixAgent also exposes a liveness probe on `:8111`, but that endpoint answers 200 while the service is still starting, so the status probe deliberately targets `:7061/v1/models` — the port consumers actually call.
- **The gateway is `https`.** `hc_health` hardcodes `http://`, which is why `hc_health_url` exists; plain `http` against the TLS listener answers 400.

Safety

- §11.4.161: uses rootless podman via the Containers submodule
- §11.4.119: single-owner GPU — only one track boots the coder
- §12.6: 60% memory ceiling applies to service boot

Cross-references

- `docs/helix_code/INTEGRATION_PLAN.md` §3 Tasks 1.2-4
- `docs/helix_code/CLAUDE_TOOLKIT_EXTENSION.md` §1
- `docs/helix_code/RISK_ANALYSIS.md` — port collisions, GPU contention